Global Grade 10 Syllabus (All Subjects)

Note: This syllabus is designed to cover core topics commonly taught in Grade 10 globally. It includes sciences, mathematics, languages, social sciences, ICT, and Islamiyat.

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1. English Language & Literature

Objectives: Reading comprehension, grammar, writing, literature analysis, vocabulary development, speaking & listening.

Topics:

1. **Reading & Comprehension**
    - **Understanding short stories, essays, and passages.**
    - **Identifying main ideas, supporting details, and inferences.**
    - **Summarizing and paraphrasing texts.**
2. **Grammar & Vocabulary**
    - **Parts of speech (nouns, verbs, adjectives, adverbs, prepositions, conjunctions).**
    - **Tenses (present, past, future, continuous, perfect, perfect continuous).**
    - **Direct and indirect speech.**
    - **Active and passive voice.**
    - **Punctuation and capitalization.**
    - **Common idioms, phrases, and synonyms/antonyms.**
3. **Writing Skills**
    - **Paragraph writing (descriptive, narrative, expository).**
    - **Essay writing (argumentative, descriptive, reflective).**
    - **Formal and informal letters.**
    - **Reports and emails.**
    - **Story writing.**
4. **Literature**
    - **Poetry: comprehension, poetic devices (simile, metaphor, imagery, symbolism).**
    - **Short stories: analysis of characters, plot, theme, and moral.**
    - **Drama: reading and interpretation of selected plays.**
    - **Novel study: themes, characters, and plot summary.**
5. **Speaking & Listening**
    - **Oral presentations.**
    - **Debates and group discussions.**
    - **Listening comprehension exercises.**

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2. Mathematics

Objectives: Problem-solving, analytical thinking, and application of mathematical concepts.

Topics:

1. **Algebra**
    - **Quadratic equations.**
    - **Factorization and expansion.**
    - **Polynomials.**
    - **Linear inequalities.**
2. **Geometry**
    - **Coordinate geometry (distance, midpoint, slope).**
    - **Circles and tangents.**
    - **Triangles and congruency.**
    - **Pythagoras theorem.**
3. **Trigonometry**
    - **Trigonometric ratios (sin, cos, tan, sec, cosec, cot).**
    - **Trigonometric identities.**
    - **Heights and distances.**
4. **Probability & Statistics**
    - **Probability rules and applications.**
    - **Mean, median, mode.**
    - **Standard deviation.**
5. **Calculus (basic introduction, optional in some countries)**
    - **Differentiation basics.**
    - **Simple applications (rate of change).**
6. **Mensuration**
    - **Surface area and volume of 3D shapes.**
7. **Vectors & Matrices (optional)**
    - **Basic operations.**

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3. Physics

Objectives: Understanding laws of nature, motion, energy, and applications.

Topics:

1. **Mechanics**
    - **Laws of motion (Newton's laws).**
    - **Work, energy, and power.**

    - **Momentum and collisions.**
    - **Gravitation and motion of planets.**
2. **Heat & Thermodynamics**
    - **Temperature and heat.**
    - **Thermal expansion.**
    - **Laws of thermodynamics.**
3. **Waves & Sound**
    - **Wave properties.**
    - **Sound and its characteristics.**
    - **Doppler effect.**
4. **Light & Optics**
    - **Reflection and refraction.**
    - **Lenses and mirrors.**
    - **Human eye and optical instruments.**
5. **Electricity & Magnetism**
    - **Current, voltage, resistance (Ohm's law).**
    - **Series and parallel circuits.**
    - **Magnetic fields and electromagnetism.**
6. **Modern Physics**
    - **Atomic structure.**
    - **Radioactivity.**
    - **Nuclear energy basics.**

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4. Chemistry

Objectives: Understanding chemical properties, reactions, and real-life applications.

Topics:

1. **Atomic Structure**
    - **Atoms, electrons, protons, neutrons.**
    - **Isotopes and electronic configuration.**
2. **Chemical Bonding**
    - **Ionic, covalent, metallic bonds.**
    - **Polar and non-polar molecules.**
3. **Stoichiometry**
    - **Mole concept.**
    - **Balancing chemical equations.**
4. **States of Matter**
    - **Solid, liquid, gas.**
    - **Gas laws (Boyle, Charles, Ideal Gas Law).**

5. **Acids, Bases, & Salts**
    - **pH, neutralization.**
    - **Properties and reactions.**
6. **Organic Chemistry Basics**
    - **Hydrocarbons (alkanes, alkenes, alkynes).**
    - **Alcohols, acids, and esters.**
7. **Chemical Reactions**
    - **Oxidation and reduction.**
    - **Combustion, decomposition, displacement.**
8. **Environmental Chemistry**
    - **Pollution, green chemistry basics.**

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5. Biology

Objectives: Understanding living organisms, systems, and environment.

Topics:

1. **Human Anatomy & Physiology**
    - **Circulatory, respiratory, digestive, excretory systems.**
    - **Nervous system and endocrine glands.**
2. **Genetics & Evolution**
    - **Mendelian genetics.**
    - **DNA structure and function.**
    - **Principles of evolution.**
3. **Ecology & Environment**
    - **Ecosystems, food chains, food webs.**
    - **Biodiversity and conservation.**
4. **Microbiology**
    - **Bacteria, viruses, fungi.**
    - **Human diseases and immunity.**
5. **Plant Biology**
    - **Photosynthesis.**
    - **Transport systems in plants.**
    - **Reproduction in plants.**

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6. Social Studies / History / Geography

Objectives: Knowledge of global history, geography, civics, and social issues.

Topics:

1. **History**
    - **World Wars I & II causes, events, outcomes.**
    - **Colonialism and independence movements.**
    - **Cold War basics.**
2. **Geography**
    - **Continents, oceans, climate zones.**
    - **Plate tectonics, earthquakes, volcanoes.**
    - **Human geography: population, urbanization.**
3. **Civics / Political Science**
    - **Forms of government (democracy, monarchy, dictatorship).**
    - **Human rights and global institutions.**
    - **Citizenship and responsibilities.**
4. **Economics (basic)**
    - **Supply and demand.**
    - **Inflation and trade.**
    - **National income basics.**

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7. Computer / ICT

Objectives: Develop digital literacy and basic programming understanding.

Topics:

1. **Introduction to Computers**
    - **Hardware and software.**
    - **Input/output devices.**
2. **Programming Basics**
    - **Algorithms and flowcharts.**
    - **Simple coding (Python, Java, or pseudocode).**
3. **Internet & Networking**
    - **Browsers, email, online safety.**
    - **Networking fundamentals.**
4. **Productivity Tools**
    - **Word processing, spreadsheets, presentations.**
5. **Cybersecurity**
    - **Passwords, data protection, malware awareness.**

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8. Islamiyat / Religious Studies

Objectives: Understanding Islamic teachings, morals, and history.

Topics:

1. **Beliefs & Principles**
    - **Six pillars of faith.**
    - **Five pillars of Islam.**
2. **Life of Prophet Muhammad (SAW)**
    - **Early life, prophethood, key events.**
3. **Important Prophets**
    - **Stories and lessons from Musa (AS), Dawood (AS), Isa (AS), Ibrahim (AS).**
4. **Islamic Ethics**
    - **Honesty, patience, charity, respect.**
5. **Worship Practices**
    - **Prayer, fasting, zakat, hajj.**
6. **Contemporary Issues**
    - **Islamic perspective on modern challenges.**

⍰ This Grade 10 global syllabus covers all major subjects, Islamiyat included, and can be applied universally.